Madan Bhandari Memorial Academy Nepal
Madan Bhandari College of Engineering
Pokhara University

Lesson Plan

Name of Faculty: **Er. Pankaj Shah**
Subject : **Big Data Technologies (Elective Subject)**

Discipline : **BE(Computer)**
Year : IV
Semester : **I**

Credit Hour : **3-0-0**

Course Objectives

1. **This chapter deals with the introduction to Big Data, defining what actually big data means.**
2. **The limitations of the traditional database, which led to the evolution of Big Data, are explained, and insight into Big Data key concepts is delivered.**
3. **A comparative study is made between big data and traditional database giving a clear picture of the drawbacks of the traditional database and advantages of big data.**
4. **The 3 Vs of big data (Volume, Velocity, and Variety) that distinguish it from the traditional database are explained.**
5. **With the evolution of big data, which are no longer limited to the structured data only.**
6. **The different types of human-and machine –generated data – that is, structured, semi-structured, and unstructured – that can be handled by big data are explained.**
7. **The chapter expands to show the various stages of big data life cycle starting from data generation, acquisition, preprocessing, integration, cleaning, transformation, analysis, and visualization to make business decisions.**

Week	Day	Theory	Session (hours)	Teaching Method	Practical
1	1	**Unit 1: Introduction to the world of Big Data** 1.1. Understanding Big Data 1.2. Evolution of Big Data 1.3. Failure of Traditional Database in Handling Big Data	**2**	**Slide**	**Lab 1: To download Python and install it and again to download PyCharm and install it.** i. To run Hello world to check whether the python and PyCharm are installed successfully.
	2	**Unit 1: Introduction to the World of Big Data** 1.4. Data Mining vs. Big Data 1.5. 3Vs of Big Data 1.6. Sources of Big Data	**2**	**Slide**	
2	3	**Unit 1: Introduction to the World of Big Data**	**2**	**Slide/ Board**	**Lab 2: To know about the Basic Python Application such as: Syntax,**

		1.7. Different Types of Data 1.7.1. Structured Data 1.7.2. Unstructured Data 1.7.3. Semi-Structured Data 1.8. Big Data Infrastructure			**data types and variables in Python programming.** i. To run the program to show the image in PyCharm using python. ii. To make the directory in Project and to import image in the directory.
	4	**Unit 1: Introduction to the World of Big Data** 1.9. Big Data Life Cycle 1.9.1. Big Data Generation 1.9.2. Data Aggregation 1.9.3. Data Preprocessing 1.9.4. Data Integration 1.9.5. Data Cleaning 1.9.6. Data Reduction 1.9.7. Data Transformation 1.9.8. Big Data Analytics 1.9.9. Visualizing Big Data	**2**	**Slide**	
3	5	**Unit 1: Introduction to the World of Big Data** 1.10. Big Data Technology 1.10.1. Challenges Faced by Big Data Technology. 1.10.2. Heterogeneity and Incompleteness 1.10.3. Volume and Velocity of the Data. 1.10.4. Data Storage 1.10.5. Data Privacy 1.11. Big Data Applications	**2**	**Board**	**Lab 3 : To know about the Image Enhancement and Sharpening** i. Implementing the python program to resize the image. ii. Implementing the python program to make the image digital negative. iii. To improve the quality of the image using gray scale.
	6	**Unit 1: Introduction to the World of Big Data** 1.12. Big Data Use Cases 1.12.1. Health Care 1.12.2. Telecom 1.12.3. Financial Services	**2**	**Board**	
4	7	**Unit 2: Big Data Storage Concepts** 2.1. Cluster Computing 2.2. Distribution Models	**2**	**Board**	**Lab 4: To know about different shapes and their sized and to generate them.**

<table>
<tr><td></td><td>8</td><td>Unit 2: Big Data Storage Concepts
2.3. Distributed File System
2.4. Relational and Non-Relational Databases
2.5. Scaling Up and Scaling Out Storage</td><td>2</td><td>Board/ IDE</td><td>i. To develop different shapes and manipulate their sizes using python programming.</td></tr>
<tr><td rowspan="2">5</td><td>9</td><td>Unit 3: NoSQL Database
3.1. Introduction to NoSQL
3.2. Why NoSQL
3.3. CAP Theorem</td><td>2</td><td>Board/ IDE</td><td rowspan="2">Lab 5: To know about the different operations on videos.
i. To run videos using python programming.
ii. To generate different frames and sizes of videos using python.</td></tr>
<tr><td>10</td><td>Unit 3: NoSQL Database
3.4. ACID
3.5. BASE
3.6. Schemaless Databases</td><td>2</td><td>Board/ IDE</td></tr>
<tr><td rowspan="2">6</td><td>11</td><td>Unit 3: NoSQL Database
3.7. NoSQL (Not Only SQL)
3.7.1. NoSQL vs. RDBMS
3.7.2. Features of NoSQL Databases
3.7.3. Types of NoSQL Technologies</td><td>2</td><td>Board/ IDE</td><td rowspan="4">Lab 6: To Implement different operations using python
i. To change the color of videos using python programming</td></tr>
<tr><td>12</td><td>Unit 3: NoSQL Databases
3.7.4. NoSQL Operations.
3.8. Migrating from RDBMS to NoSQL</td><td>2</td><td>Classroom</td></tr>
<tr><td rowspan="2">7</td><td>13</td><td>Test Taken</td><td>2</td><td></td></tr>
<tr><td>14</td><td>Unit 4: Processing, Management Concepts, and Cloud Computing
4.1. Data Processing
4.2. Shared Everything Architecture
4.3. Shared-Nothing Architecture</td><td>2</td><td>Board/ IDE</td></tr>
<tr><td>8</td><td>15</td><td>Unit 4: Processing, Management Concepts, and Cloud Computing
4.4. Batch Processing
4.5. Real-Time Data Processing</td><td>2</td><td>Board/ IDE</td><td>Lab 7: To Implement Object Detection using python programming
i. To create python program for object detection.</td></tr>
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		4.6. Parallel Computing 4.7. Distributed Computing			
	16	**Unit 4: Processing, Management Concepts, and Cloud Computing** 4.8. Big Data Virtualization 4.9. Cloud Computing Types 4.10. Cloud Services	**2**	**Board**	
9	17	**Unit 4: Processing, Management Concepts, and Cloud Computing** 4.11. Cloud Storage 4.12. Cloud Architecture	**2**	**Board/ IDE**	**Lab 8 : To implement Pattern Recognition system** i. To develop a python program to create a pattern for study. ii. To develop a python program to know the overall Pattern designs.
	18	**Unit 5: Driving Big Data with Hadoop Tools and Technologies** 5.1. Apache Hadoop 5.2. Hadoop Storage 5.3. Hadoop Computation	**2**	**Board/ IDE**	
10	19	**Unit 5: Driving Big Data with Hadoop Tools and Technologies** 5.4. Hadoop 2.0	**2**	**Slide/IDE**	**Lab 10: To Demonstrate the segment of Images** i. To develop a python program for creating segmentation of an image.
	20	**Unit 5: Driving Big Data with Hadoop Tools and Technologies** 5.5. HBASE 5.6. Apache Cassandra 5.7. SQOOP 5.8. Flume	**2**	**Slide**	
11	21	**Unit 5: Driving Big Data with Hadoop Tools and Technologies** 5.9. Apache Avro 5.10. Apache Pig 5.11. Apache Mahout 5.12. Apache Oozie	**2**	**Board/ IDE**	**Lab 11 : To know about the segmentation of a video.** i. To develop a python program to create a segmentation of a video.
	22	**Unit 5: Driving Big Data with Hadoop Tools and Technologies** 5.13. Apache Hive 5.14. Hive Architecture	**2**	**Board/ IDE**	
	28	**Test**	**2**		**Test + Viva(Lab + Theory)**
	29	**Model Question Revision**	**2**		